

36T TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2069 Bhadra

Exam.	Regular / Back		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF & Microwave Circuits, Systems & Devices (EG785EX) (Elective II)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary formulas and smith chart are attached herewith.
- ✓ Assume suitable data if necessary.

1. Classify microwave frequency bands and state their major applications. Describe how microwave transmission lines are different from the conventional low frequency transmission lines. [4+4]
2. Describe microwave radiation hazards based on the radiation fields. [8]
3. Describe rectangular waveguide based on modes on propagation and other critical parameters. [8]
4. Describe why S-parameter is important in microwave network analysis. Using a two-port network derive S-parameters. [4+6]
5. What is double-stub tuner? Assuming a load of $75 + j75$ ohm is connected to a 50-ohm transmission line, find the lengths and spacing for a two-stub impedance matching system with three-eighths wavelength separation between the stubs. [3+15]
6. Design an amplifier to attain maximum gain at 4.0 GHz using a GaAs FET having following S-parameters: $S_{11} = 0.72\angle -116^\circ$, $S_{12} = 0.03\angle 57^\circ$, $S_{21} = 2.60\angle 76^\circ$ and $S_{22} = 0.73\angle -54^\circ$. Consider the characteristic impedance, $Z_0 = 50$ Ohm. [18]
7. Write short notes (Any TWO)
 - a. E-plane tee against H-plane tee
 - b. PROBE-coupling against LOOP-coupling
 - c. Microstrips
 - d. Two-cavity klystron

[2 x 5]

Supplied Formulas:

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2 |S_{12}| |S_{21}|},$$

$$\Delta = (S_{11}S_{22}) - (S_{12}S_{21}),$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{21}S_{12}|}$$

$$\Gamma_s = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1},$$

$$\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}, \text{ where}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2,$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2,$$

$$C_1 = S_{11} - \Delta S_{22}^*, \text{ and}$$

$$C_2 = S_{22} - \Delta S_{11}^*$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{|S_{22}|^2 - |\Delta|^2}$$

$$C_s = \frac{(S_{11} - \Delta S_{22}^*)^*}{|S_{11}|^2 - |\Delta|^2}$$

$$R_L = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2}$$

$$R_s = \frac{|S_{12}S_{21}|}{|S_{11}|^2 - |\Delta|^2}$$

$$G_{T_{\max}} = \left(\frac{1}{1 - |\Gamma_s|^2} \right) |S_{21}|^2 \left(\frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2} \right)$$